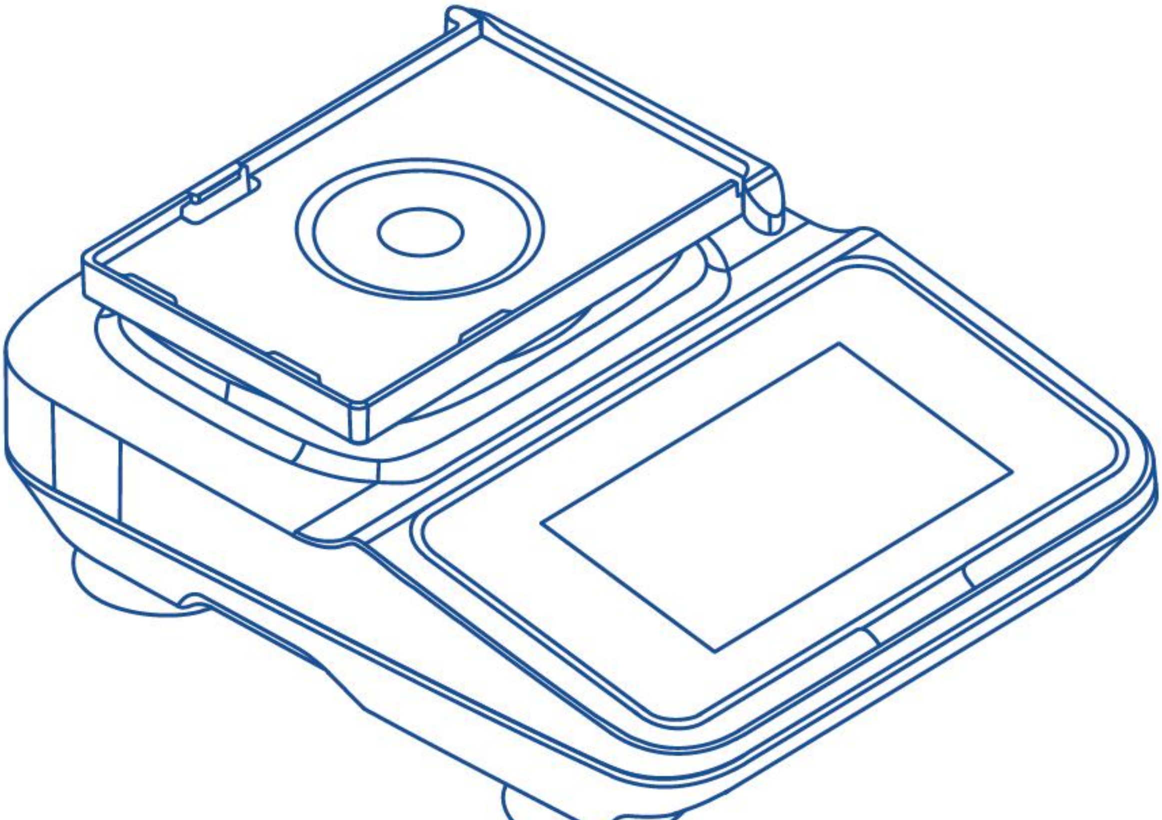
VORTEX MIXER
智能涡旋仪

OPERATION MANUAL
操作手册



QUALITY
质量好

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价格省

SUPPORT
服务安心

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3. Safety Instructions

3.1 Operating Environment

3.1.1 Indoor use only.

3.1.2 Altitude ≤ 2000m.

3.1.3 Applicable ambient temperature range: 5-40℃.

3.1.4 Applicable relative humidity range: ≤80%.

3.1.5 Power supply applicable to select product models.

3.1.6 Adequate ventilation equipment must be installed indoors.

3.1.7 Ensure there is no vibration or airflow in the surrounding area that affects performance.

3.1.8 Ensure the surrounding air contains no conductive dust, explosive gases, or corrosive gases.

3.2 Safety Tips

3.2.1 Please read this manual carefully before using the device for the first time.

3.2.2 The vortex mixer may only be operated by trained and authorized personnel.

3.2.3 Equipment maintenance must only be performed by our company or our authorized agents.

3.2.4 The following materials are strictly prohibited in the instrument:

— Flammable and explosive materials;

— Strongly reactive chemical materials;

— Toxic, radioactive substances, or pathogenic microorganisms.

3.2.5 Only qualified maintenance personnel using appropriate tools may perform repair operations on the vortex mixer system.

3.2.6 If the instrument operator encounters situations not covered in this manual, please contact our company or our authorized agents for guidance on the correct handling procedure.

3.2.7 Use accessories provided by our company whenever possible. If the user uses other accessories, our company will not be liable for any unintended consequences.

3.2.8 The instrument must be inspected and maintained at specified intervals.

3.3 Precautions

3.3.1 Do not plug/unplug the power cord or toggle the power switch with wet hands!

3.3.2 Do not plug/unplug the power cord while the instrument is powered on!

3.3.3 Do not maintain or clean the instrument while it is powered on!

3.3.4 Do not fill sample containers beyond two-thirds capacity for vortexing. To prevent liquid spillage, sample containers must have caps!

3.3.5 Do not install the instrument on uneven, swaying, or vibrating work surfaces!

3.3.6 Ensure gaskets or fixtures are properly installed to prevent them from flying out during operation!

3.3.7 The vortexed sample load must not exceed the instrument's maximum capacity!

4. Device Introduction

4.1 Product Overview

The QuanLab V5 Pro Vortex Mixer features a high-strength zinc alloy housing frame with anti-slip silicone pads. Our proprietary structural vibration attenuation technology enhances operational stability, ensuring the machine maintains high-precision speed accuracy even during extended high-speed operation, with a rated lifespan exceeding 30,000 hours. The control method replaces traditional knobs and buttons with an advanced 4.3-inch touchscreen interface that provides users with an innovative and smooth vortexing experience. The manual control interface and multi-program step operation settings are simple, supporting storage of up to 10 programs for one-touch recall, with run curves providing clear program status at a glance. The machine's exterior design and touchscreen UI adopt a fresh, natural design philosophy, adding a touch of delight to your routine lab work!

4.2 Model and Technical Specifications

4.2.1 Product Model: V5 Pro

4.2.2 Input Power: AC100-250V 50/60Hz

4.2.3 Output Power: 10W

4.2.4 Motor: Brushless DC Motor

4.2.5 Control Method: CLK Speed Regulation

4.2.6 Operation Modes: Touch / Continuous / Programmable

4.2.7 Speed Range (rpm): 300-3000

4.2.8 Vortex Mode: Orbital

4.2.9 Display: 4.3-inch Touchscreen

4.2.10 Timer Function: 99h 59min 59s

4.2.11 Orbital Diameter (mm): 4.5

4.2.12 Weight (kg): 4.3

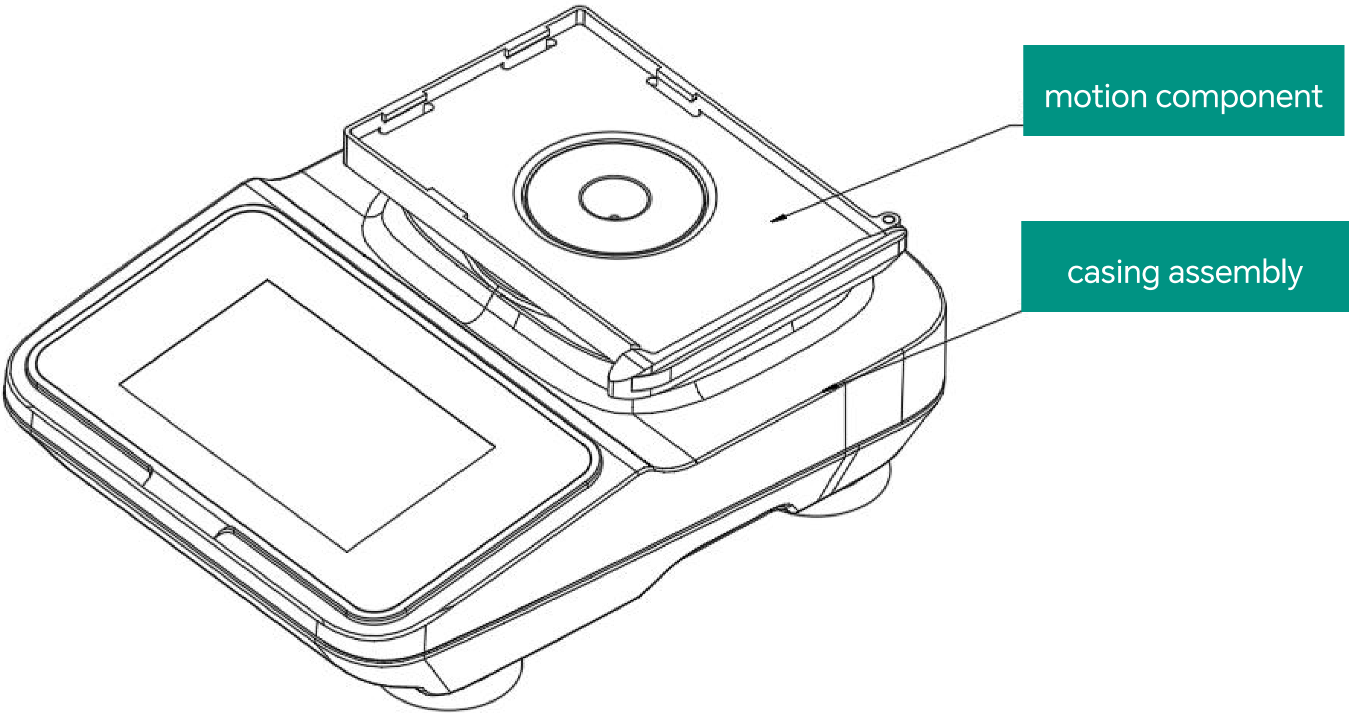
4.2.13 Dimensions (L×W×H; mm): 225×168×85

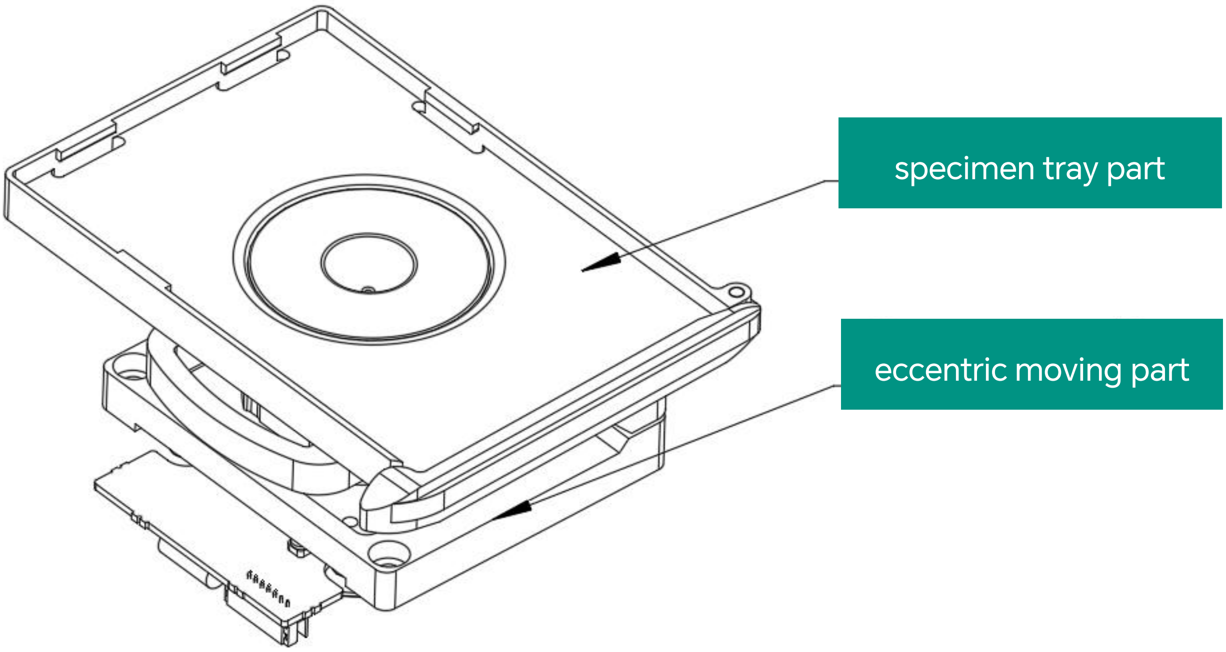
4.2.14 Noise at Max Speed: ≤65dB(A)

4.2.15 Body Material: Zinc Alloy

4.2.16 Compatible Modules: 0.6 / 1.5 / 2 / 5 / 10 / 15 / 50 mL centrifuge tubes, 96-well plates, 384-well plates

4.3 Device Components





4.4 Function Introduction

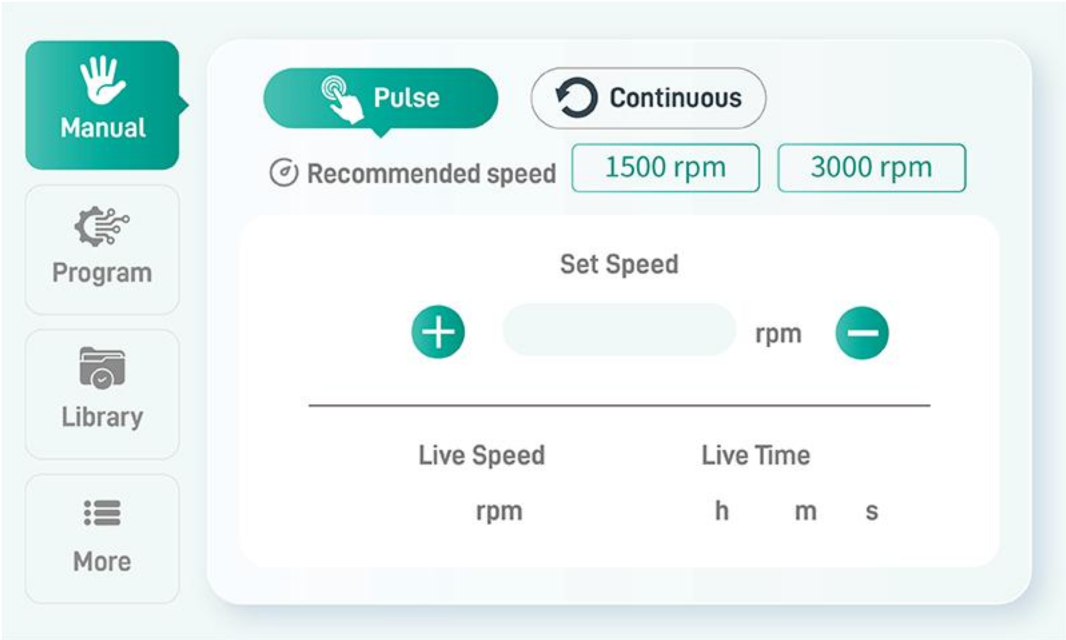
4.4.1 Startup Page

On startup, a loading progress bar is displayed. Once loading is complete, it jumps to the manual settings page.

4.4.2 Manual Settings

Manual settings are divided into Touch mode and Continuous mode.

4.4.2.1 Manual Settings - Touch Mode



Simply set the speed. Range: 300-3000 rpm. A prompt will appear if the range is exceeded.

Real-time Speed: The real-time rotational speed of the motor.

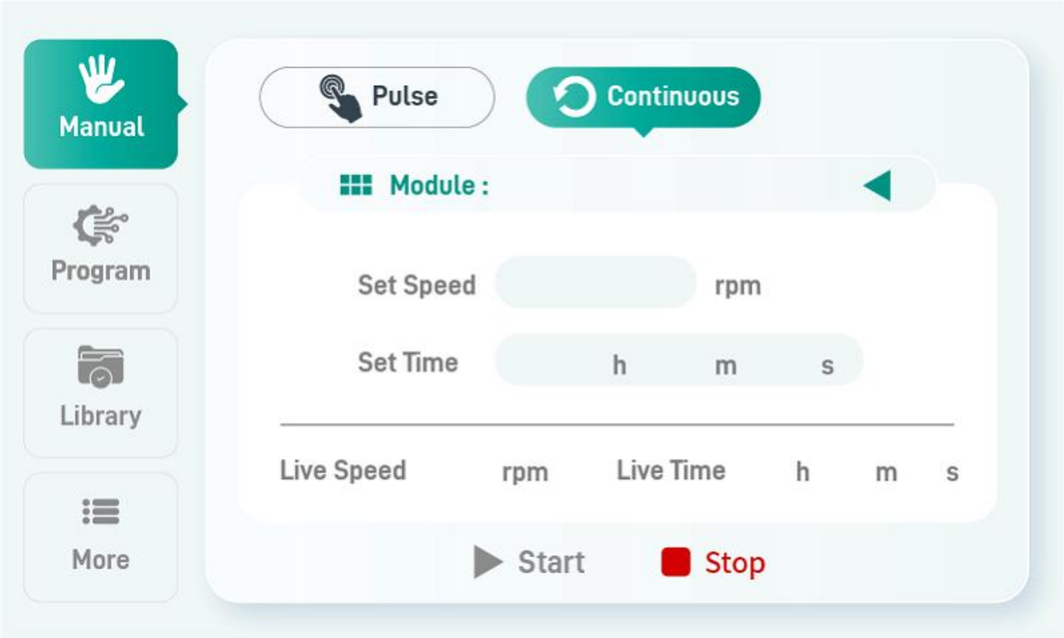
Real-time Time: Count-up timer, starts counting when the motor starts rotating and stops when operation ceases.

+ : Tap once; the set speed value increases by 100.

- : Tap once; the set speed value decreases by 100.

Note: The set speed is retained after power loss; upon repowering, the previous value is restored.

4.4.2.2 Manual Settings – Continuous Mode



Set speed. Range: 300 to the upper speed limit for the corresponding module. A prompt will appear if the range is exceeded.

Set time in hours, minutes, and seconds. Maximum: 99:59:59.

Real-time Speed: The real-time rotational speed of the motor.

Real-time Time: Count-up timer, starts counting when the motor starts rotating and stops when operation ceases.

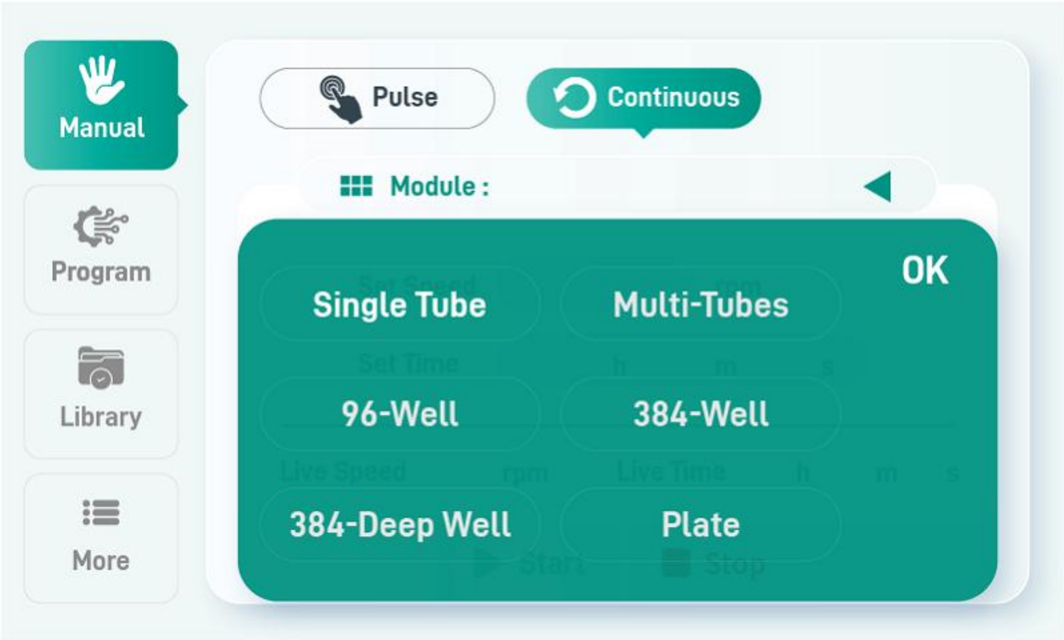
Start: Press the Start button to start the motor; the Stop button light turns off.

Stop: When the countdown ends and the motor stops, the Stop button lights up and the Start button turns off.

Note: ① Pressing the Stop button at any time will stop the motor.

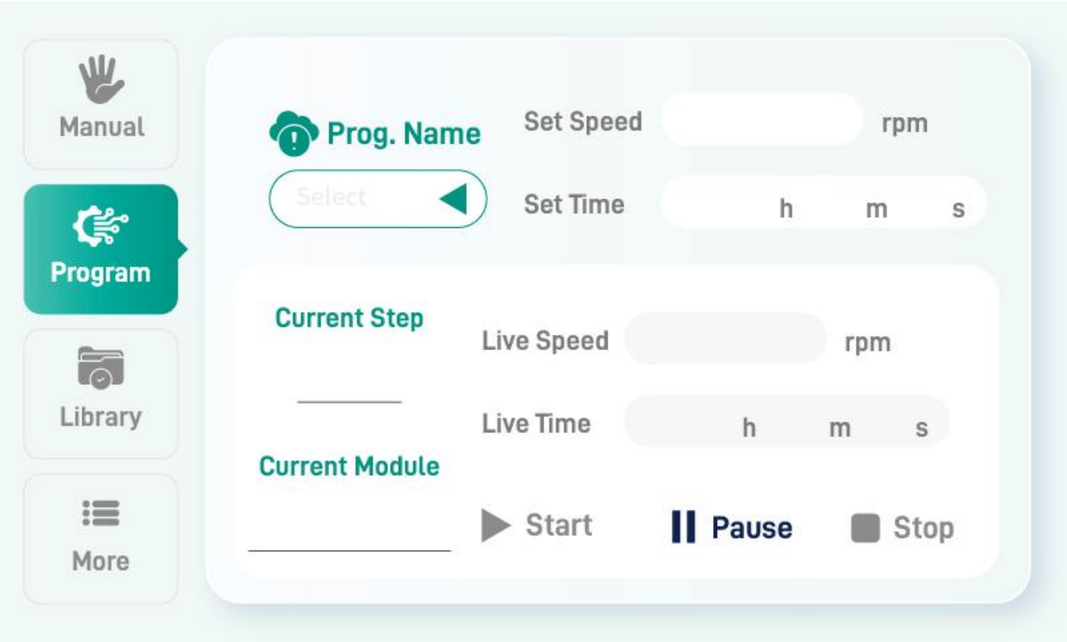
② The displayed set speed is the recommended speed value for the module model (i.e., the upper limit, configured in the system settings page). The user may change it, but it must be less than this recommended value.

4.4.2.3 Current Module Selection



Tap the corresponding module to select it; the highlighted cursor indicates selection. Tap "Apply Module" to return to Manual Settings – Continuous Mode, where "Set Speed" will display the recommended speed value for that module.

4.4.3 Program Control



Set Speed: The target motor speed, not configurable here; this field directly displays the step parameters set on the Program Library page.

Set Time: The target motor time (H:MM:SS), not configurable here; this field directly displays the step parameters set on the Program Library page.

Real-time Speed: The real-time rotational speed of the motor.

Real-time Time: Count-up timer, starts counting from motor start and stops when the target time is reached. This process repeats for each step.

Start: Press the Start button to start the motor; the Stop button light turns off.

Stop: When the countdown ends and the motor stops, the Stop button lights up and the Start button turns off.

Pause: Press the Pause button to stop the motor; the timer holds its value. When the motor runs again, timing continues without resetting.

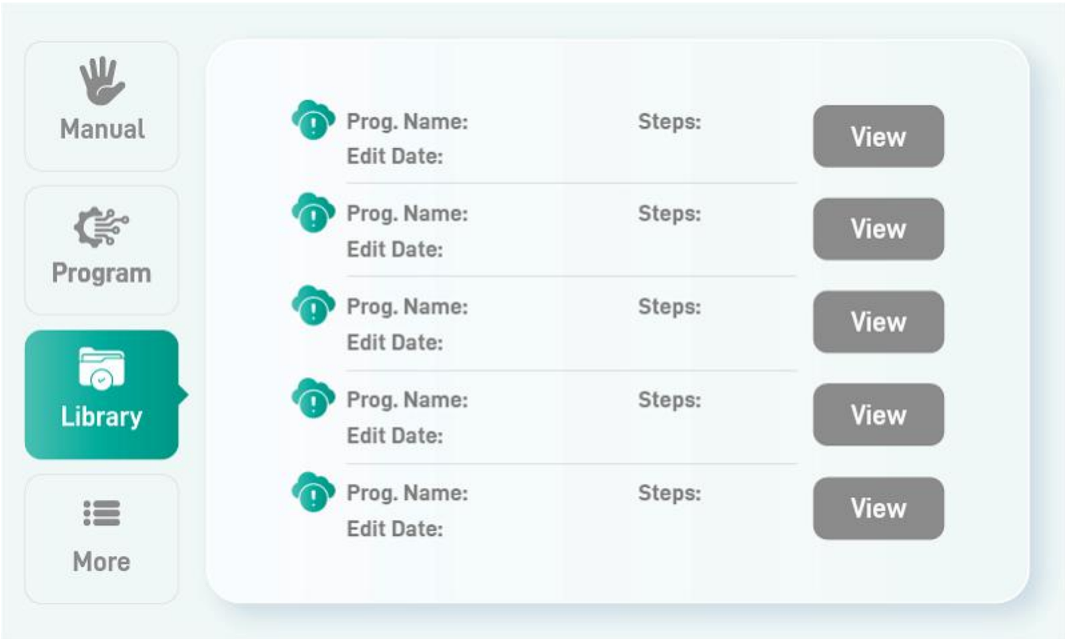
Program Name: Displays the currently selected program. Two selection methods:

① Input directly on the current page to select. ② Select from the Program Library page.

Current Step: Displays the currently running program step.

Note: Pressing the Stop button at any time will stop the motor.

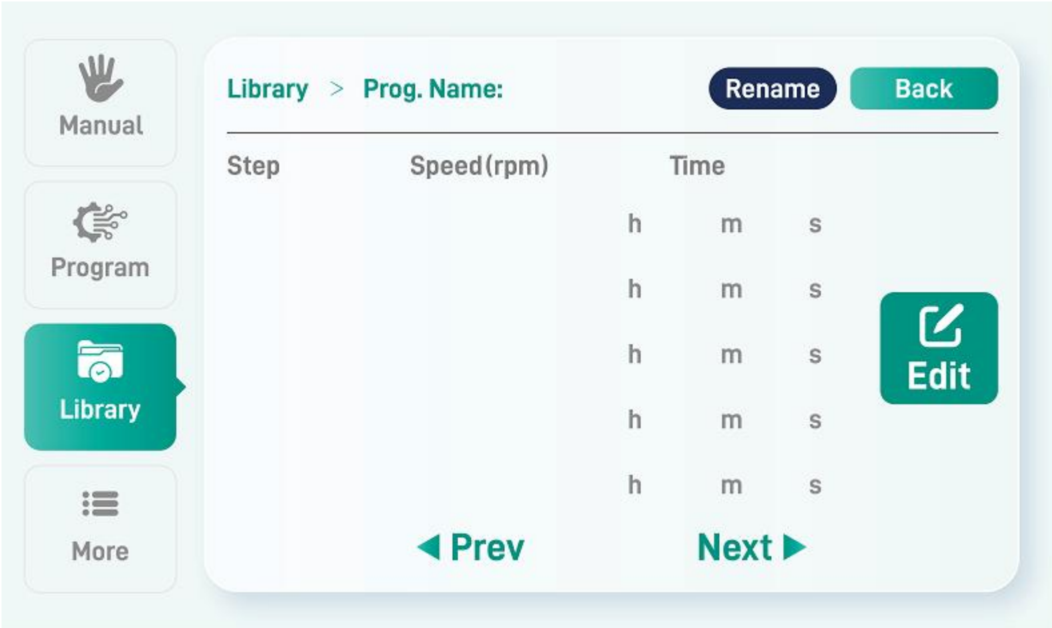
4.4.4 Program Library



On the Program Library page, you can select a program to run and view detailed program steps. There are 5 programs, each capable of containing up to 10 steps. Configured programs are retained after power loss and restored upon repowering.

Tap the View button to see detailed data for the current program.

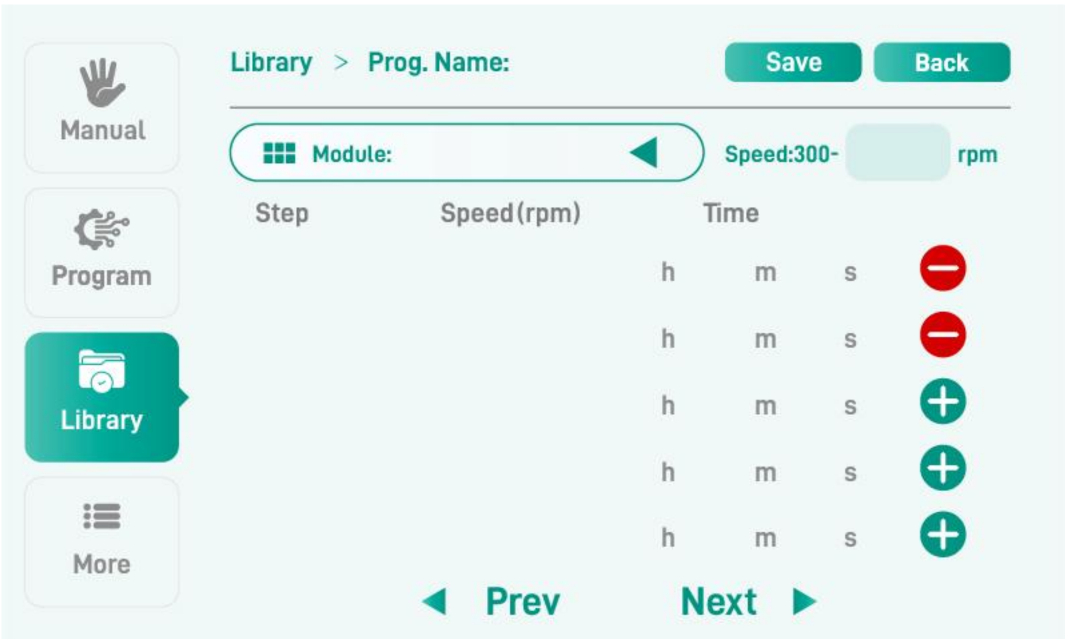
4.4.4.1 Program Step View



1. Tap the Edit button to go to the program editing page, where you can edit the current program's data.

2. Tap Rename to change the current program name.

4.4.4.2 Program Editing Page



(1) There are five plus (+) buttons on the far right of the page.

① When a button is pressed, the text box in the corresponding row becomes editable.

② When a button is released, the data in the corresponding row is deleted, and subsequent program steps shift upward.

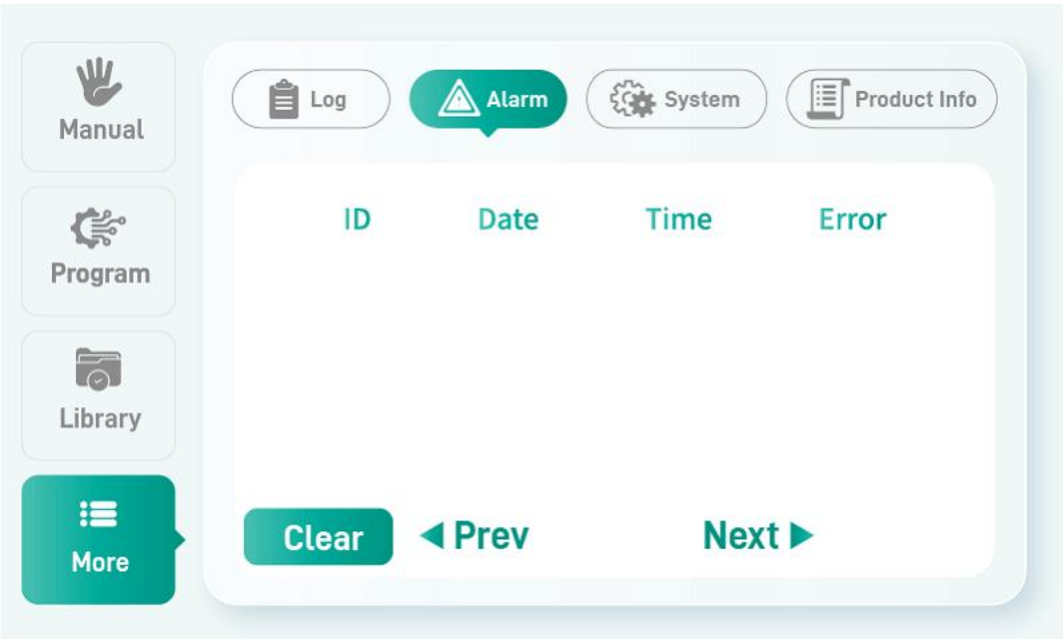
③ The 10 program steps must be added sequentially; if the previous button is not pressed, the next button cannot be activated.

(2) You can flip through pages using Previous Page and Next Page. A single program can save up to 10 steps.

(3) After editing, tap the Save button to successfully save the current program step parameters.

Note: Tapping the "Cancel" button returns you to the Program Library page.

4.4.5 Log Page



(1) Five columns: Program Name, Start Time (H:MM:SS), End Time (H:MM:SS), Run Mode (Touch, Continuous, Programmable), and Details.

(2) Tap the Details icon to enter the Log Details page, which displays more detailed and complete information about that log entry.

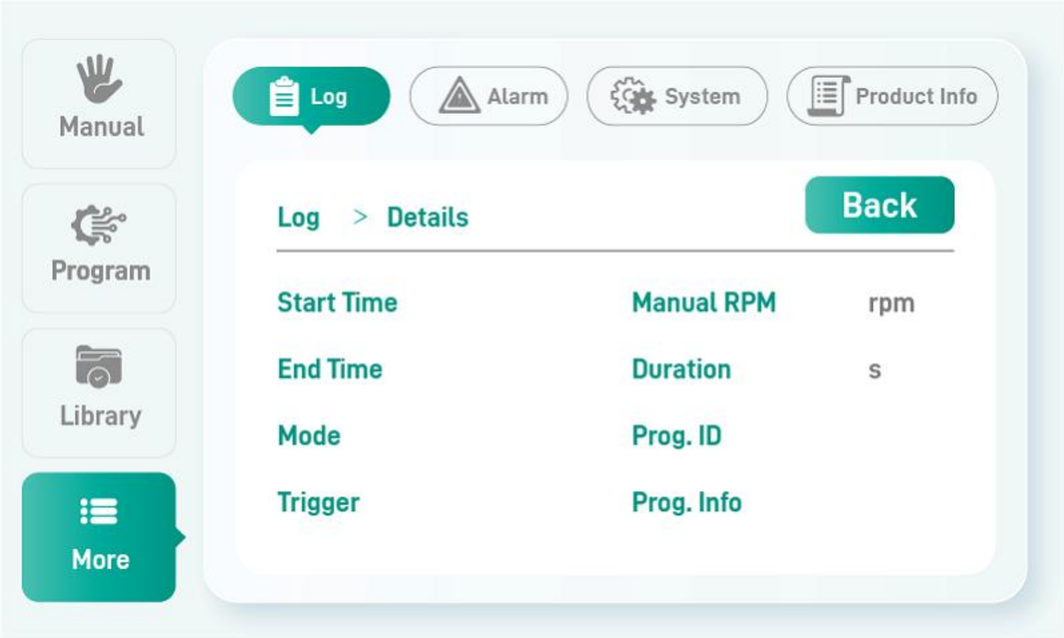
(3) Up to 20 log entries can be saved. Navigate using the Previous/Next Page buttons.

(4) Logs also have power-loss retention functionality.

(5) Tap the "Clear" button to erase all log information. (Note: This process may take some time.)

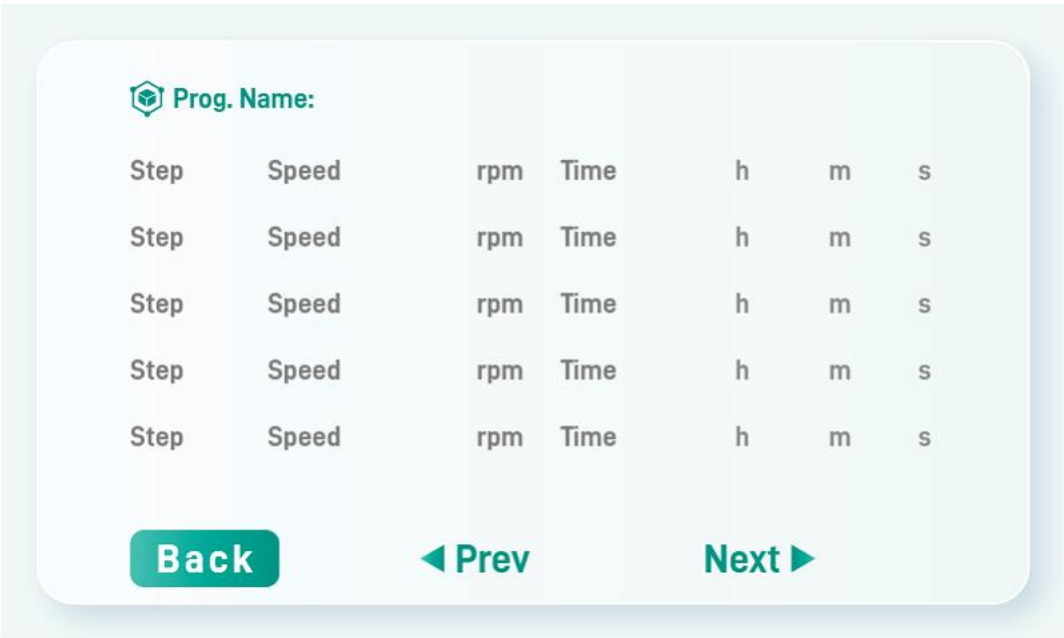
Important Note: After clicking the "Clear" button, log erasure will take some time. Restart the V5 Pro Vortex Instrument once the operation is complete.

4.4.5.1 Log Details Page

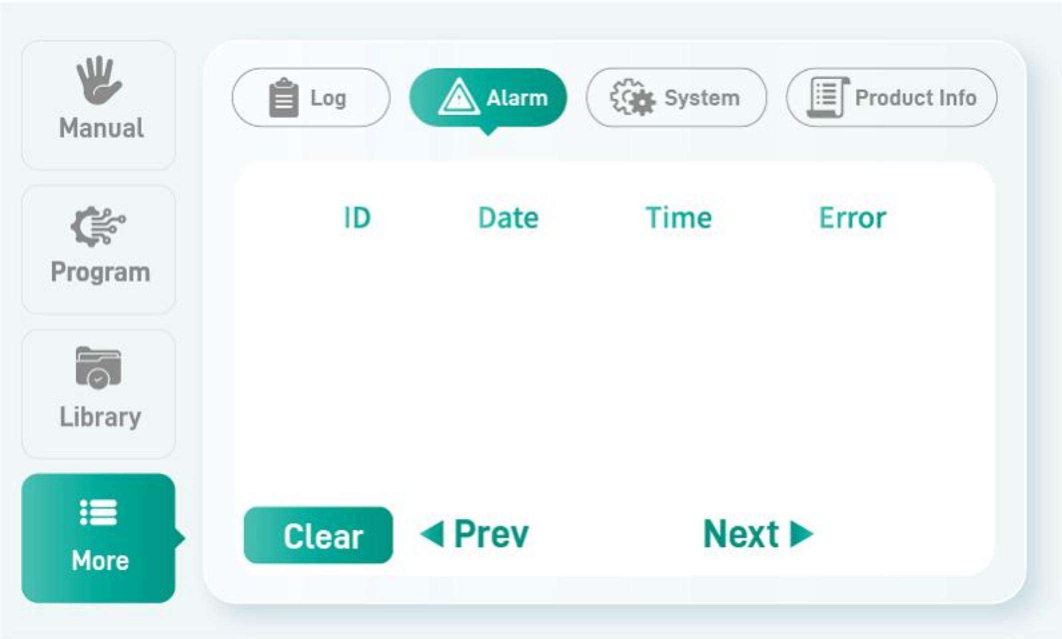


Tapping the Details icon enters the Log Details page, displaying detailed information about that log entry.

4.4.5.2 Program Details Page



4.4.6 Alarm Page

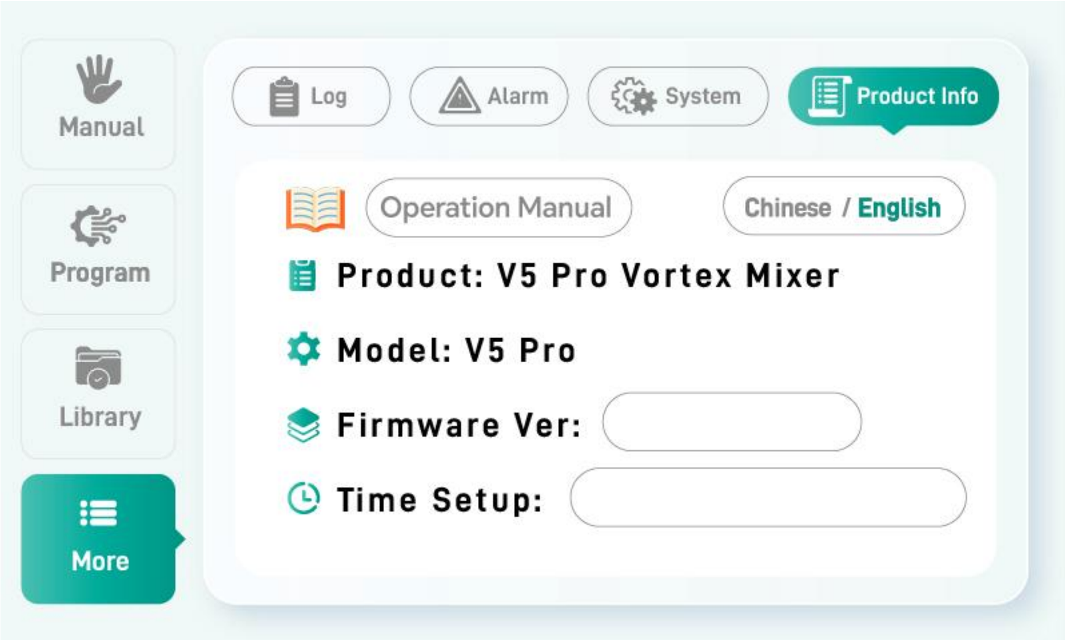


(1) Displays alarm information. Up to 20 entries can be saved. Navigate using the Previous/Next Page buttons.

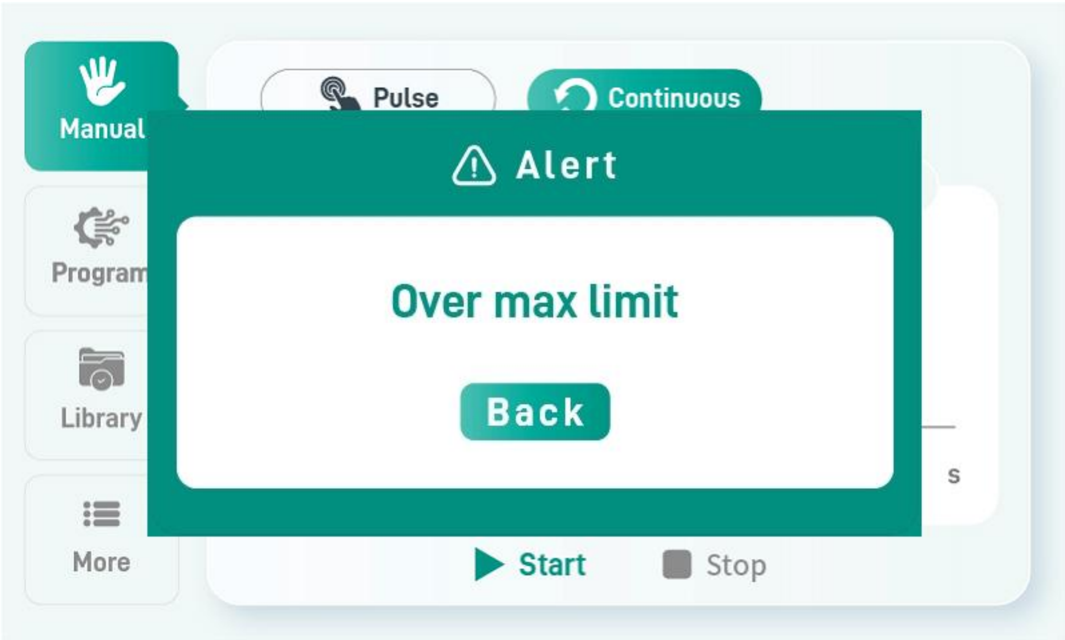
(2) Alarm information also has power-loss retention functionality.

(3) Tap the "Clear" button to erase all alarm information. (Note: This process may take some time.)

4.4.7 Product Information

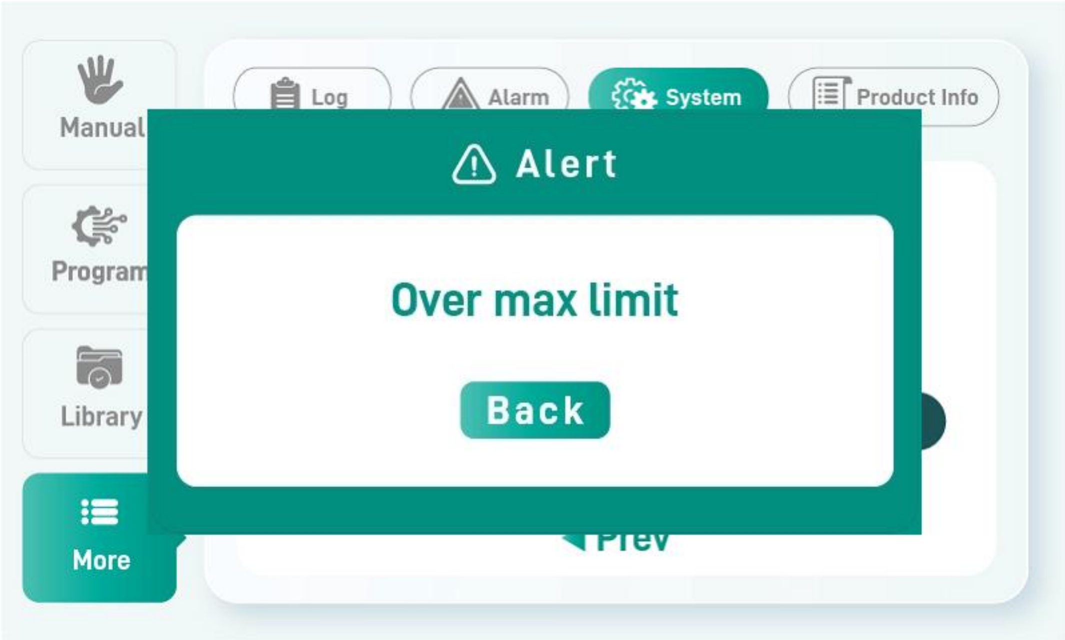


4.4.8 Speed Exceeds Upper Limit



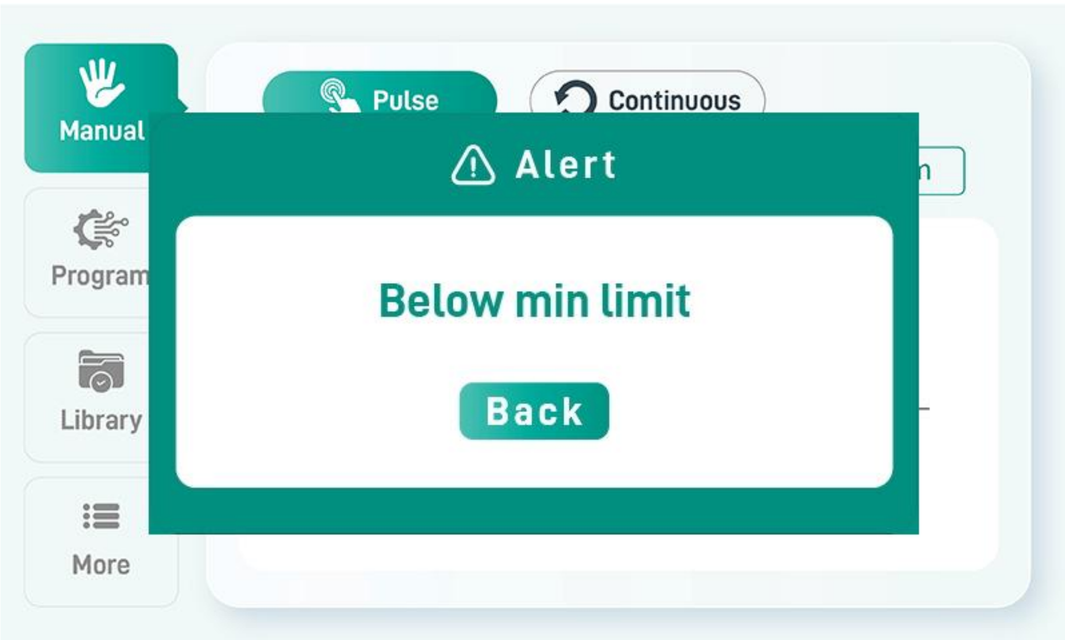
In Continuous mode, if the set speed exceeds the recommended speed for the corresponding module model, a pop-up prompt will appear.

4.4.9 Speed Exceeds Absolute Upper Limit



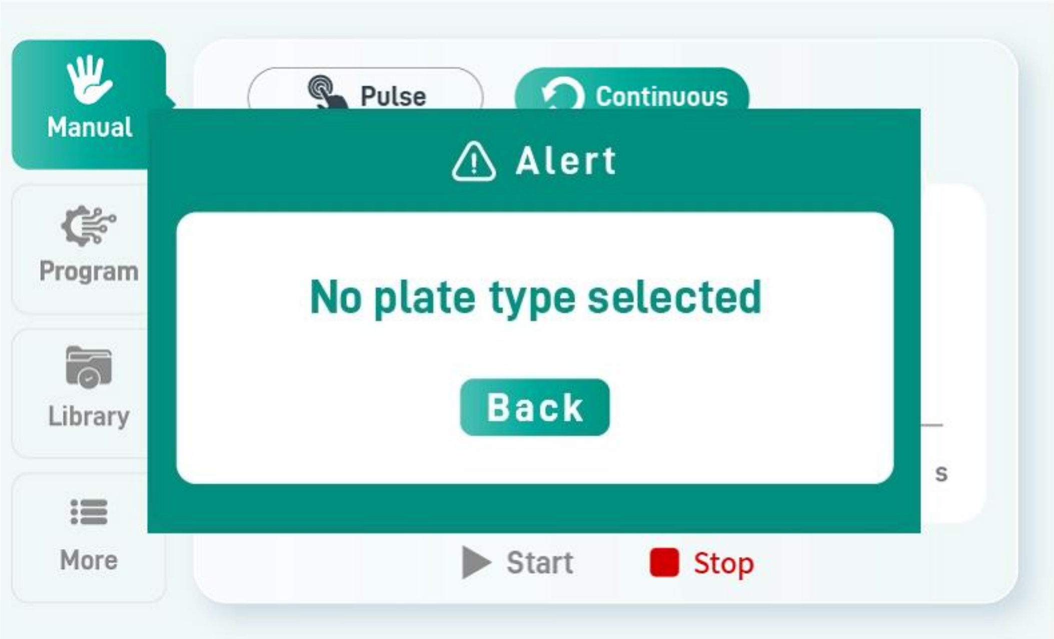
On the System Settings page, if the module's corresponding speed exceeds the absolute upper limit, a pop-up prompt will appear.

4.4.10 Speed Below Lower Limit



In Touch mode and Continuous mode, if the set speed is below the lower limit, a pop-up prompt will appear.

4.4.11 Module Model Not Recognized

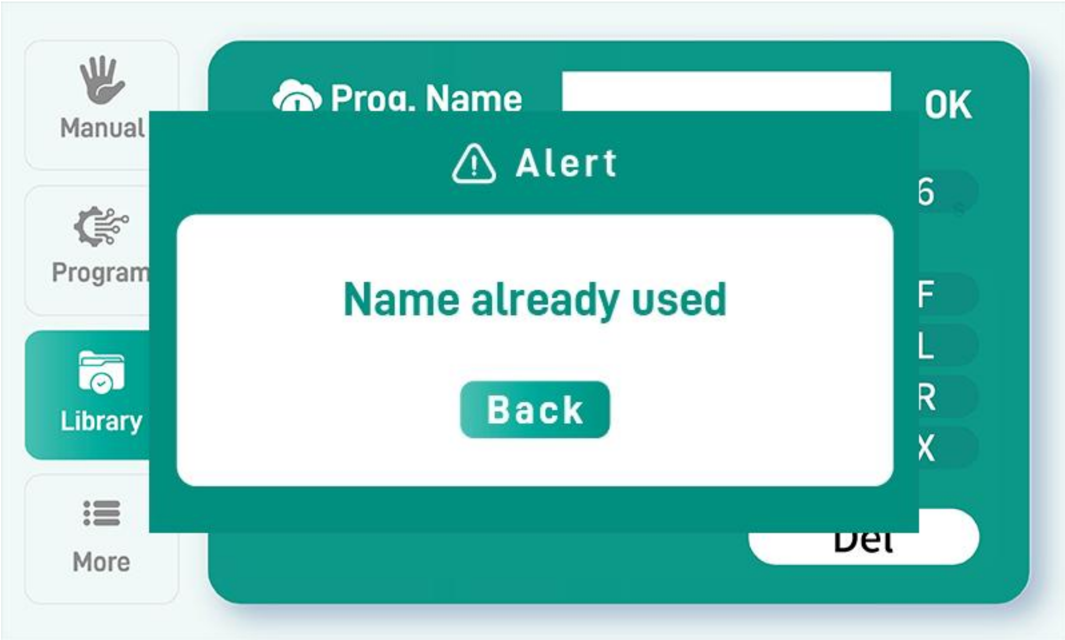


In Continuous mode, if Run is pressed without selecting a module, a pop-up prompt will appear.

4.4.12 Invalid Program Name

If the entered program name is empty, a pop-up prompt will appear.

4.4.13 Duplicate Program Name



If the entered program name duplicates an existing one, a pop-up prompt will appear.

4.5 Function Description

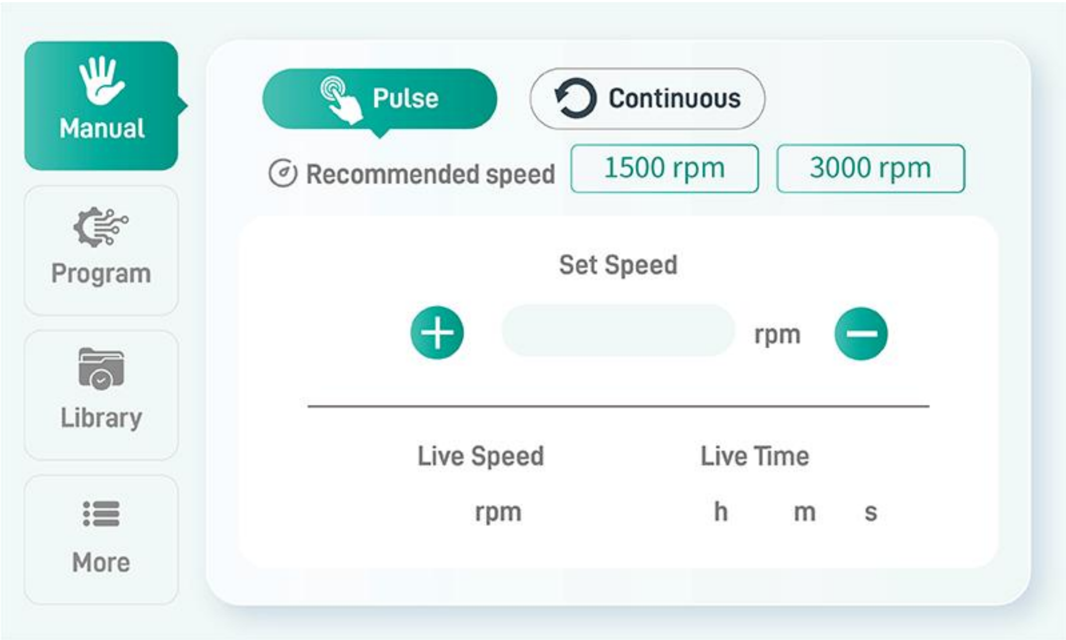
4.5.1 Three Operation Modes

(1) Touch Mode: Triggered when pressure reaches the set value. Only the speed needs to be set; the run time is controlled by the user (lifting the tube reduces pressure to a certain level, stopping the motor).

(2) Continuous Mode: Triggered by pressing the Run button. Requires selecting the module model and setting speed and time. Stopped by pressing the Stop button or when the set time elapses.

(3) Programmable Mode: Triggered by pressing the Run button. Requires selecting a program on the Program Library page with corresponding step parameters (target speed and run time). Stopped by pressing the Stop button or when all steps are completed.

4.5.2 4.5.2 Touch Mode



4.5.2.1 Prerequisites

① ① Operation Mode: Manual Settings - Touch Mode

② Set Speed: 300-3000 (rpm)

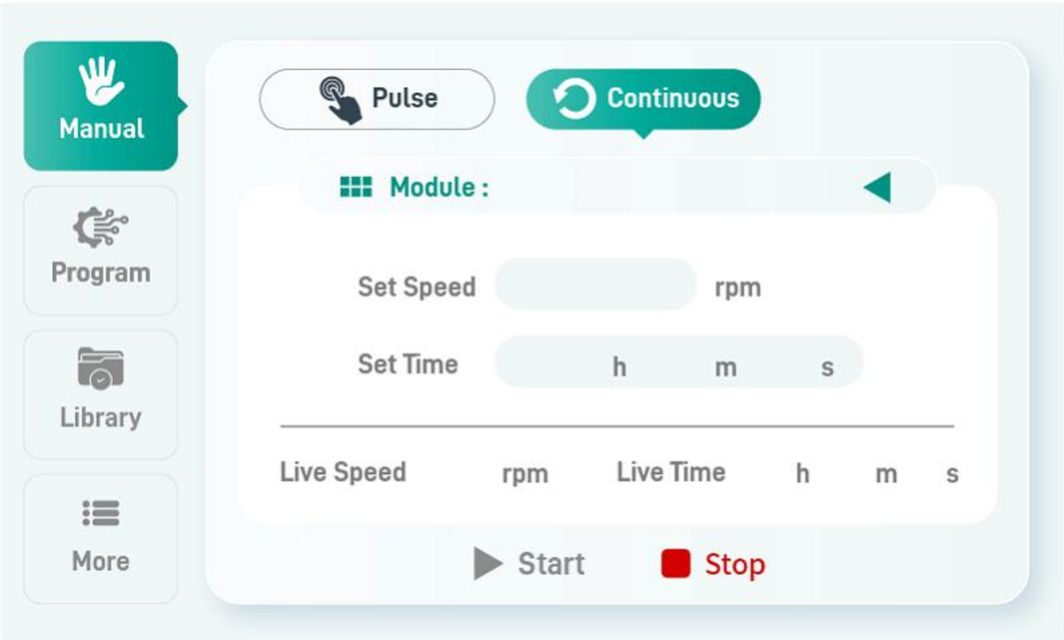
4.5.2.2 Operation Procedure

① Place the tube in the designated position and press firmly (Note: a certain amount of pressure must be maintained throughout operation). When the pressure exceeds the set value, operation begins and the motor rotates. (If a buzzer is installed, it sounds once.)

② When the reagent reaches the required state, lift the tube; operation stops and the motor halts. (If a buzzer is installed, it sounds three times consecutively.)

Note: If after powering on, the motor does not move when adjusting speed, an alarm will sound (buzzer rapidly beeps five times) and an alarm entry (Motor Fault) will be logged.

4.5.2 4.5.3 Continuous Mode



4.5.3.1 Prerequisites

① ① Operation Mode: Manual Settings - Continuous Mode

② ② Tap "Apply Module" and select the module model

③ Set Speed: Must be lower than the recommended speed

③ ④ Set Time

4.5.3.2 Operation Procedure

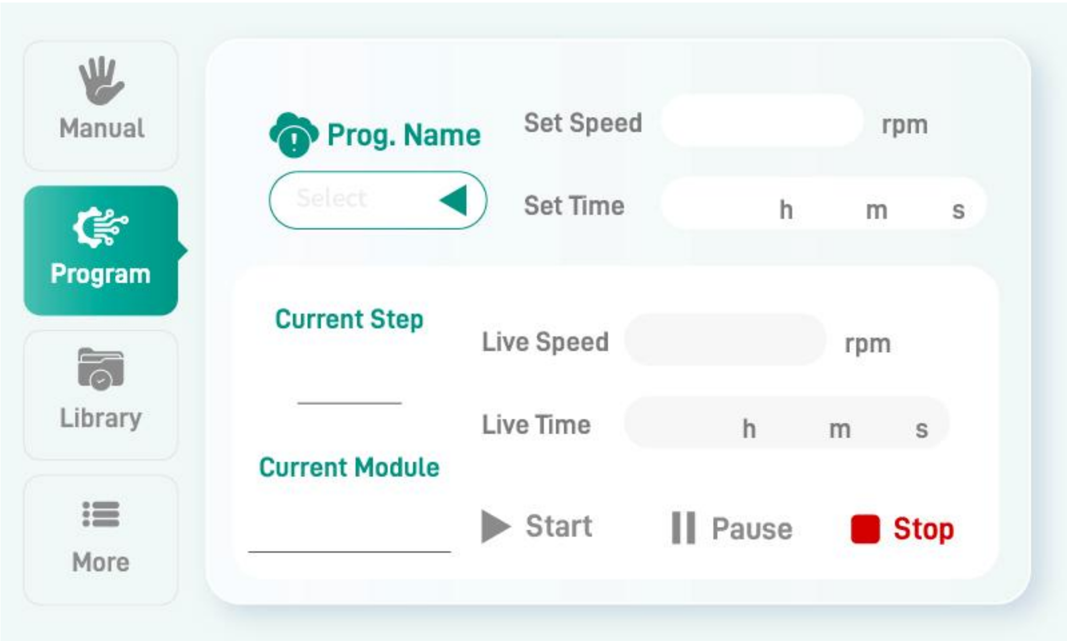
① ① Press Run; the motor starts rotating. (The buzzer sounds once.)

② When the set operating time is reached, the motor stops rotating. (The buzzer sounds three times consecutively.)

Note: ① During operation, pressing the Stop button will also stop the motor.

② If after powering on, the motor does not move when adjusting speed, an alarm will sound (buzzer rapidly beeps five times) and an alarm entry (Motor Fault) will be logged.

4.5.3 4.5.4 Programmable Mode



4.5.4.1 Prerequisites

① ① Operation Mode: Program Control

② ② Select the program to run on the Program Library page

③ Return to the run page to view the current program number and current step; the set speed and time are already displayed.

4.5.4.2 Operation Procedure

① ① Press Run; the motor starts rotating.

② When the set operating time for the current step is reached, the program automatically switches to the next step, simultaneously updating the current step, set speed, and set time in real time.

③ When all steps of the program have been executed, the motor stops rotating.

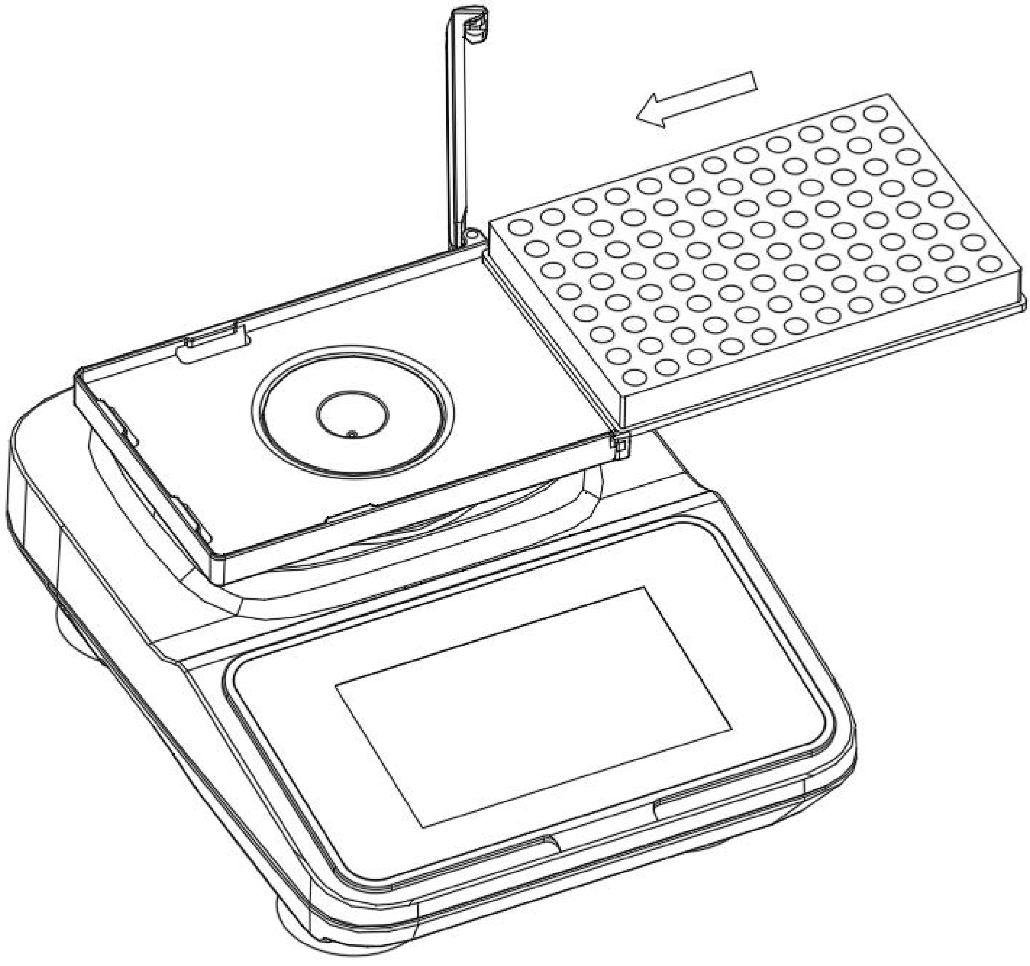
Note: ① During operation, pressing the Stop button will stop the motor.

② When all steps are completed, the current step, set speed, and set time will revert to the data corresponding to Step 1.

③ If after powering on, the motor does not move when adjusting speed, an alarm will sound (buzzer rapidly beeps five times) and an alarm entry (Motor Fault) will be logged.

4.6 Operation Instructions

4.6.1 Installation of Standard Gaskets and Plate Fixtures



Open the clamping plate latch by sliding it to the right. Push the plate fixture into the sample mounting plate from right to left all the way in (Note: it must snap under the left-side notch!). Close the clamping plate latch. The plate accessory installation is complete!

5. Maintenance

5.1 Routine Maintenance

5.1.1 Surface Cleaning

5.1.1.1 Do not attempt to clean the instrument while the power cord is plugged in or the power switch is in the ON position.

5.1.1.2 Immediately after use, wipe the instrument surface with a soft cloth or alcohol swab to prevent sample residue (e.g., chemical reagents, biological fluids).

5.1.1.3 Stains can be cleaned with a neutral detergent; avoid corrosive solvents (e.g., strong acids, strong bases).

5.1.2 Vortex Head Maintenance

5.1.2.1 Periodically inspect the rubber pad (vortex head) for wear, cracking, or contamination; promptly clean with alcohol or replace as needed.

5.1.2.2 If the rubber pad slips, moisten the surface with a small amount of deionized water (avoid organic solvents that may corrode the rubber).

5.1.3 Power Cord Inspection

5.1.3.1 Check the power cord for damage or poor contact to avoid short circuit risks.

5.2 Periodic Maintenance

5.2.1 Mechanical Component Lubrication

5.2.1.1 For frequent use, apply a small amount of high-temperature grease (e.g., silicone grease) to motor bearings and transmission components every 6 months to reduce friction noise.

5.2.1.2 Caution: Power must be disconnected and the housing opened first. It is recommended that this be performed by professional personnel.

5.2.2 Internal Dust Removal

Use compressed air or a soft brush to remove internal dust to prevent it from affecting heat dissipation or circuit performance.

5.2.3 Tightening Screws

Inspect and tighten loose screws, especially those on the vortex head and base mounting components.

5.2.4 Disassembly and Transport

Do not strike fixtures with sharp objects. Prevent impacts during transport, disassembly, and assembly to avoid cracks, scratches, or external damage to fixtures and the main unit.

6. Troubleshooting

6.1 Device Won't Start

6.1.1 Possible Causes: Power failure, damaged switch, burnt-out motor.

6.1.2 Troubleshooting Steps:

① Check whether the power outlet and wiring are functioning properly. ② Use a multimeter to test switch continuity; replace damaged switches. ③ If the motor does not respond and there is a burning smell, the motor needs replacement (contact the manufacturer or professional repair personnel).

6.2 Abnormal Noise

6.2.1 Possible Causes: Bearing lacking lubrication, loose components, detached rubber pad.

6.2.2 Troubleshooting Steps: ① Re-secure or replace the rubber pad. ② Lubricate the bearings (if the noise persists, bearing replacement may be necessary).

7. Usage Precautions

7.1 Load Limit

7.1.1 Avoid prolonged overload operation (e.g., excessively heavy or

oversized containers) to prevent motor damage.

7.1.2 Recommended container weight: Generally not exceeding 0.5 kg (including fixtures).

7.2 Environmental Requirements

7.2.1 Keep away from humid, high-temperature, or corrosive gas environments.

7.2.2 Turn off the power after use to reduce standby power consumption.

7.3 Calibration Recommendations

7.3.1 Calibrate the speed with a tachometer once a year to ensure the displayed value matches the actual speed (especially important for experiments with high precision requirements).

7.4 Professional Maintenance Recommendations

7.4.1 If the instrument cannot be repaired through simple troubleshooting, contact the manufacturer or an authorized service center; avoid self-disassembly to prevent secondary damage.

7.4.2 Retain the purchase receipt and warranty card; free periodic maintenance services are provided.

8. Module List

No.	Sales Item No.	Sales Product Name	Sales Model	Sales Specification
1	QT-QV1	Module V1 (0.65mL centrifuge tubes ×30)	--	1 piece/box
2	QT-QV2	Module V2 (1.5/2mL centrifuge tubes × 30)	--	1 piece/box
3	QT-QV3	Module V3 (5mL centrifuge tubes ×16)	--	1 piece/box
4	QT-QV4	Module V4 (10/15mL centrifuge tubes ×5)	--	1 piece/box
5	QT-QV5	Module V5 (50mL centrifuge tubes ×2)	--	1 piece/box
6	QT-QV6	Module V6 (0.65/1.5/2mL mixed centrifuge tubes ×30)	--	1 piece/box
7	QT-QV7	Module V7 (Flat plate type)	--	1 piece/box

9. After-Sales Service

9.1 Warranty Service

9.1.1 3-Year Whole-Unit Warranty: Free repair or replacement of core components such as the motor and mainboard (for non-human-caused damage).

9.1.2 1-Year Screen/Touch Module Warranty: Free replacement for touchscreen unresponsiveness or display issues caused by normal use.

9.2 Repair Service

9.2.1 Rapid Response: Technical response to after-sales issues within 24 hours on working days, with a solution provided within 48 hours.

9.2.2 Nationwide Warranty: Supports mail-in repair or on-site service (Shanghai area supports on-site service within 2 working days).

9.2.3 Backup Unit Support: If the repair cycle exceeds 5 working days, a free backup unit can be requested.

9.3 Parts Replacement

9.3.1 Consumables Discount: During the warranty period, wear parts such as rubber pads and adapters are eligible for 60% off replacement.

9.3.2 Lifetime Maintenance: After the warranty expires, only cost-price fees are charged, with official OEM parts provided.

9.4 Technical Support

9.4.1 Professional Consultation: Free professional technical support for experiment method optimization, troubleshooting, and more.

9.4.2 Remote Diagnostics: Rapid problem-solving guidance via app or video call.

9.5 Return and Exchange Policy

7-Day Quality Issue Replacement: Direct replacement with a new unit for non-human-caused performance failures within 7 days of receipt.

9.6 Value-Added Services

Free Calibration: One official performance calibration per year (requires return shipment or on-site appointment).

Device Warranty Card

Product Information

Product Name: Intelligent Vortex Mixer

Model: ____________

Serial No.: ____________

Purchase Date: ___________

Warranty Policy

Warranty Period: Free warranty service for 36 months from the date of purchase.

Warranty Coverage: Performance failures caused by material or manufacturing defects (e.g., motor abnormality, circuit board failure, sensor malfunction, etc.). Free repair or replacement of damaged parts during the warranty period (for non-human-caused damage).

Warranty Proof: This warranty card and a valid proof of purchase (invoice/receipt) are required.

Disclaimer (Not Covered Under Warranty)

The following situations require paid repair:

Human-caused damage (e.g., dropping, liquid infiltration, improper operation, etc.).

Use or maintenance not in accordance with the manual requirements, or unauthorized modification or disassembly of the product.

Damage caused by natural disasters or force majeure (e.g., fire, flood, lightning strike, etc.).

Failures caused by unauthorized third-party repairs.

Normal wear and tear or consumables (e.g., tube pads, adapters, and other wear parts).

Warranty Service Process

Fault Reporting:

Contact the customer service hotline/email, describe the fault symptoms, and provide the product serial number and proof of purchase.

After customer service confirmation, you will be informed of the repair drop-off or mail-in procedure.

Repair Processing:

After receiving the product, an engineer will inspect it and provide feedback within 5-7 working days.

Faults within warranty coverage are repaired free of charge; for non-warranty issues, repair proceeds after the user confirms the cost.

Product Return: After repair is completed, the product will be shipped back to the user's designated address (free shipping during the warranty period).

Contact Information

Service Hotline: 400-025-2120

Email: quanlab@lab-direct.com

Official Website: www.quan-lab.com

Repair Center Address: Building 21, Baijiatong, No. 388 Shengrong Road, Pudong New District, Shanghai

Precautions

Please keep this warranty card and proof of purchase properly; they cannot be reissued if lost.

Friendly Reminder: Regular cleaning and maintenance can extend the life of the device. Please refer to the manual for instructions.